Separating forcing and low-frequency feeding in a signed Navier–Stokes tail budget

September 27, 2026

Abstract

For a smooth finite-energy forced Navier–Stokes evolution, we bound the absolute time integrals of direct force work and low-frequency feeding in the high-frequency enstrophy balance. The bounds use kinetic energy, total energy dissipation, a fixed cutoff, and derivatives of the force; they spend none of the high-frequency viscous dissipation. Consequently, for the forced singularities constructed in [1], the accumulated tail self-stretching minus even the full tail dissipation is unbounded at every fixed cutoff. The full-field obstruction is checked in Lean. The Fourier-tail extension below is an analytic proof, not a formalized projector theorem. No unforced regularity estimate is obtained.

1 An exact split and finite allowances

Let u, p, f solve

$$\partial_t u + (u \cdot \nabla)u = -\nabla p + \nu \Delta u + f, \qquad \nabla \cdot u = 0, \qquad \nu > 0,$$

on $\mathbb{R}^3 \times [a, T)$, where $0 < L = T - a < \infty$. Assume smoothness and, on each compact slab $[a, b] \subset [a, T)$, $u \in C^1([a, b]; H^m(\mathbb{R}^3))$ and $f, \nabla p \in C([a, b]; H^m(\mathbb{R}^3))$ for every integer $m \geq 0$. These hypotheses justify the following strong energy identities and integrations by parts. Write

$$E(t) = \|u(t)\|_2^2, \quad Y(t) = \|\nabla u(t)\|_2^2 = \|\omega(t)\|_2^2, \quad \omega = \operatorname{curl} u.$$

Suppose

$$E(t) \leq M, \quad \|f(t)\|_2 \leq F_0, \quad \|\Delta f(t)\|_2 \leq F_\Delta \qquad (a \leq t < T).$$

All constants in this note are independent of the shorter endpoint t. The kinetic-energy identity gives

$$\int_a^t Y(s)\, \mathrm{d}s \leq B, \qquad B = \frac{E(a)}{2\nu} + \frac{L\sqrt{M} F_0}{\nu}. \tag{1}$$

Indeed, $E'/2 = -\nu Y + \langle u, f \rangle$ and $E(t) \geq 0$.

Fix an angular Fourier cutoff $K \geq 0$, and set

$$u_L = \mathbf{1}_{\{|\xi| < K\}}(D)u, \quad u_H = u - u_L, \quad \omega_H = \operatorname{curl} u_H, \quad S_H = \tfrac{1}{2}(\nabla u_H + \nabla u_H^{\mathsf{T}}).$$

These are orthogonal L^2 projections commuting with derivatives. Define the physical tail quantities

$$Z_H = \tfrac{1}{2}\|\omega_H\|_2^2, \qquad D_H = \|\nabla \omega_H\|_2^2 = \|\Delta u_H\|_2^2, \qquad \Pi_H = \int_{\mathbb{R}^3} \omega_H \cdot S_H \omega_H \, \mathrm{d}x.$$

In particular, Π_H uses the high field in all three factors. With $\Pi = \int \omega \cdot S\omega$, introduce

$$P_L = \langle -\Delta u_L, u \times \omega \rangle, \quad P_H = \langle -\Delta u_H, u \times \omega \rangle, \quad X_H = P_H - \Pi_H, \quad F_H = \langle -\Delta u_H, f \rangle.$$

Then $P_L + P_H = \Pi$, and the exact tail balance is

$$Z_H' = \Pi_H + X_H - \nu D_H + F_H. \tag{2}$$

The filtered velocity is not assumed to solve an autonomous Navier–Stokes equation: X_H retains every lower-frequency input.

Proposition 1. *Let $c_K = (K^3/(6\pi^2))^{1/2}$. Under the preceding assumptions,*

$$\int_a^t |F_H(s)|\,\mathrm{d}s \le L\sqrt{M}F_\Delta, \tag{3}$$

$$\int_a^t |X_H(s)|\,\mathrm{d}s \le 3Kc_K\sqrt{M}B + K^2c_KM\sqrt{LB} =: C_L. \tag{4}$$

Neither bound uses a gradient supremum or tail dissipation of the velocity.

Proof. Integration by parts gives $F_H = -\langle u_H, \Delta f \rangle$, so $|F_H| \le \sqrt{M}F_\Delta$. This proves (3).

For the unitary Fourier convention, Cauchy–Schwarz on the ball gives

$$\|\nabla u_L\|_\infty \le Kc_K\sqrt{E}, \qquad \|\Delta u_L\|_\infty \le K^2c_K\sqrt{E}.$$

Here the gradient uses the Frobenius norm, and $c_K^2 = (2\pi)^{-3}|B_K|$. The vector and matrix bounds follow by Cauchy–Schwarz before taking a component sum.

For matrices define the trilinear form

$$b(A,B,C) = \sum_{i,j,k} A_{ji}B_{kj}C_{ki}, \qquad |b(A,B,C)| \le |A|_F|B|_F|C|_F.$$

The integrated stretching identity is $\Pi = -\int b(\nabla u, \nabla u, \nabla u)$, and the same identity holds for the smooth solenoidal field u_H. Put $A = \nabla u$, $H = \nabla u_H$ and $G = \nabla u_L = A - H$. The exact telescoping identity

$$b(A,A,A) - b(H,H,H) = b(G,A,A) + b(H,G,A) + b(H,H,G)$$

and $\|H\|_2 \le \|A\|_2$ imply

$$|\Pi - \Pi_H| \le 3\,\|\nabla u_L\|_\infty\, Y \le 3Kc_K\sqrt{M}\,Y. \tag{5}$$

Also

$$|P_L| \le \|\Delta u_L\|_\infty \|u\|_2 \|\omega\|_2 \le K^2c_KM\sqrt{Y}.$$

Now $X_H = \Pi - \Pi_H - P_L$. Integrate these two bounds and use (1) and $\int_a^t \sqrt{Y} \le \sqrt{LB}$. □

Corollary 2. *If a finite constant A satisfies*

$$\int_a^t (\Pi_H - \nu D_H)\,\mathrm{d}s \le A \qquad (a \le t < T), \tag{6}$$

then the complete gradient energy obeys

$$Y(t) \le 2Z_H(a) + 2A + 2C_L + 2L\sqrt{M}F_\Delta + K^2M.$$

In particular, a half-dissipation budget implies this conclusion, but the strictly smaller fraction is unnecessary for this fixed-cutoff gradient-energy implication.

Proof. Integrate (2), apply Proposition 1, and use $Y = 2Z_H + \|\nabla u_L\|_2^2 \le 2Z_H + K^2 M$. □

For an unforced solution, one can take $M = E(a)$ and $B = E(a)/(2\nu)$. Thus the low-frequency allowance is controlled by the datum, viscosity, cutoff and horizon. This does not prove (6): its uniform signed high-frequency history remains the open estimate. Allowing the full dissipation simply exposes its relation to the gradient-energy continuation criterion; it is not a new coercive estimate.

2 Application to the actual forced construction

The whole-space construction in [1] has smooth compactly supported forcing, bounded kinetic energy, zero initial velocity, and unbounded speed as $t \uparrow 1$. On slabs exhausting $[1/2, 1)$, all hypotheses above hold with fixed M, F_0, F_Δ.

The module `MettapediaNS.ForcedSignedBudget` in the Lean library applies directly to the construction's `CandidateProperties`. It proves that its gradient energy cannot have a uniform bound: such a bound would control the order-two energy via the forced viscous estimate, and the H^2 embedding would bound speed. It also proves the full-field signed-production obstruction directly, with force work bounded by $6\sqrt{E_0(u)}\sqrt{E_2(f)}$ in the doubled energy convention, where $E_j(v) = \|\nabla^j v\|_2^2$ sums all ordered derivative components. The constructor theorem is imported at the pinned revision `f9e8bc5b38b6e212696e8a30e3e91517af887bbd`.

Corollary 3. *For every such candidate, every fixed finite $K \ge 0$, and every $\beta \le 1$,*

$$\sup_{1/2 \le t < 1} \int_{1/2}^{t} (\Pi_H(s) - \beta\nu D_H(s))\, \mathrm{d}s = +\infty.$$

Meanwhile $\int_{1/2}^{1} |X_H|$ and $\int_{1/2}^{1} |F_H|$ are finite.

Proof. For $\beta = 1$, a finite upper bound would give a uniform gradient-energy bound by Corollary 2, contradicting the preceding forced-candidate theorem. For $\beta \le 1$, the accumulated quantity only increases because $D_H \ge 0$. The other assertions follow from Proposition 1. □

Thus the signed tail self-interaction is the term that defeats the budget; an infinite direct forcing-work allowance is not the cause. This conclusion applies to the complete constructed solution, without replacing its oscillatory corrections by a finite numerical model.

Forcing is still essential to that constructed history. With its zero datum and zero force, the kinetic-energy identity forces the smooth finite-energy solution to remain zero. In the construction the force is the smoothly extended momentum residual of the designed field (Section 10 of [1]). Removing it while keeping that field would change the equation. Restarting an unforced evolution from a later nonzero slice is a different problem, whose subsequent history is not settled by this argument. A proposed unforced budget must therefore use more than bounded forcing norms or bounded direct force work.

3 Verification boundary

The formal theorem `openAI_solution_no_uniform_signed_production_budget` quantifies over every positive viscosity and the actual imported construction. It concerns the full-field production path, identified in Lean with the spatial cubic integral, and has no frequency projector. The proof of Proposition 1 and the extension to every fixed Fourier cutoff in Corollary 3 are analytic arguments

given here. They have not been registered as Lean axioms or as kernel-checked projector theorems. No conclusion about arbitrary-data unforced regularity follows from the forced obstruction.

References

[1] OpenAI, *Finite time blowup for Navier–Stokes*, manuscript, with formalization in NavierStokesAndEuler, 2026.